

Resolution Tradeoff in Gameplay Experience, Performance, and Simulator Sickness in Virtual Reality Games

Jialin Wang¹, Rongkai Shi¹, Zehui Xiao¹, Xueying Qin² and Hai-Ning Liang^{1,*}

ABSTRACT

Higher resolution is one of the main directions and drivers in the development of Virtual Reality (VR) Head-Mounted Displays (HMDs). However, given their associated higher cost, it is unclear the benefits of having higher resolution on user experience, especially in VR games. This research aims to investigate the effects of resolution in gameplay experience and simulator sickness for VR games. To this end, we designed an experiment to collect gameplay experience, simulator sickness (SS), and player performance data with a VR First-Person Shooter game. Our results indicate that 2K resolution is an important threshold for an enhanced gameplay experience without affecting performance and increasing SS levels.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

There is intensified competition among the Virtual Reality (VR) Head-Mounted Display (HMD) manufacturers for hardware improvements. The HMD's resolution, often considered a measure of the level of detail provided by its two displays (one for each eye), can directly affect users' perception, performance, and experience when using the HMD. Compared to other aspects such as refresh rate and field-of-view, the emphasis on resolution improvements has been moving significantly more rapidly. Table 1 shows the resolution of the main consumer-level VR HMDs released in the past five years (2016 to 2021). Improving the display resolution has always been one of the drivers in the evolution of VR HMDs.

Low-resolution VR displays in an HMD may give an unclear view of the Virtual Environments (VEs) and this can lead to unclear vision and a decreased performance and experience in VR applications. For example, in games, players may not be able to recognize items (such as types of weapons and ammo) and differentiate between enemies or team players in an First-Person Shooter (FPS) game. In addition, players may not be able to observe the surroundings with clarity and enjoy the views if it is rendered in low resolution. Finally, a low-resolution application may induce Simulator Sickness (SS) [4], which remains a difficult challenge and obstacle when long exposure and rapid movements in VEs are needed [6, 5, 9, 7] and hinders the broader adoption of VR HMDs by non-regular users [8].

A high-resolution VR HMD, though it can provide more visual details of the VEs, requires a better hardware setup with higher performance, which might be costly for HMD manufacturers and users. Also, it can compete with other features, like refresh rate and frame rate. Therefore, it is worth exploring how much the resolution can benefit players during gameplay in VR environments, like games,

¹J. Wang, R. Shi, Z. Xiao, H.-N. Liang are with the Department of Computing, Xi'an Jiaotong-Liverpool University, Suzhou, China, 215028.

²X. Qin is with the School of Computer Science and Technology, Shandong University, Qingdao, China, 266237.

*Corresponding author (haining.liang@xjtlu.edu.cn).

Table 1: Resolutions of the main consumer-level VR HMDs from 2016 to 2021. The data is sorted in ascending order by 'resolution per eye'. The table shows three clear groups (1K, 2K, and 3K and above).

Device	Resolution per eye	Release date
Oculus Rift	1080×1200 (1K)	March 2016
HTC VIVE	1080×1200 (1K)	April 2016
Sony PlayStation VR	960×1080 (1K)	October 2016
FOVE	1280×1440 (1K)	January 2017
HTC VIVE Pro	1440×1600 (1K)	January 2018
Oculus Rift S	1280×1440 (1K)	March 2019
Oculus Quest 2	1832×1920 (2K)	October 2020
HP Reverb G2	2160×2160 (2K)	November 2020
Pico Neo 3	1832×1920 (2K)	May 2021
StarVR	2560×1440 (2.5K)	April 2020
HTC VIVE Pro 2	2448×2448 (2.5K)	June 2021
Varjo VR-3	2880×2720 (3K)	September 2021
PIMAX 8K	3840×2160 (4K)	February 2019
PIMAX 12K	5620×2720 (6K)	October 2021

which is an aspect that has not been explored by previous research. If a higher resolution does not help reduce SS and has no significant improvement on game experience, VR HMD manufacturers can choose display panels with lower resolution but have higher refresh rate, while VR players can choose a lower resolution setting but higher frame rate to achieve a more optimal experience. Thus, we were interested in understanding to what extent the resolution of a VR game can affect SS levels, experience, and performance.

This research aims to answer this research question by conducting a user study with a popular and highly-rated VR game, *Half-Life: Alyx*, with four resolutions (1K, 2K, 3K, and 4K). Similar to the previous study on evaluating the impact of resolution in non-VR games [1], we used an FPS game as testing VE and render resolutions with the same display. Our results show that 2K resolution is an important threshold for game experience improvement and SS reduction in VR FPS games. This paper contributes to report an empirical study investigating the effects of resolutions on gameplay experience, performance, and simulator sickness.

2 USER STUDY

The goal of this user study was to examine the effects of resolution on gameplay experience, performance, and simulator sickness. We chose a game level from *Half-Life: Alyx* as the testing VE for the experiment. Participants were required to defeat 5 enemies in the environment in 5 minutes. In short, the gameplay was a FPS task which participants were required to navigate in the scene, defeat enemies, and avoid their attacks.

The study followed a within-subjects design, where resolution was the independent variable. We collected participants' performance data and feedback after playing the selected game level from *Half-Life: Alyx* with four levels of resolution: 1096×696 (1K), 2084×1320 (2K), 3084×1956 (3K), and 3836×2432 (4K) (see Fig. 1). The order of the conditions was determined through a balanced 4×4 Latin-Square design to mitigate carry-over effects. Participants were given time to rest between two conditions.



Figure 1: Screenshots of the four resolution conditions in the experiment. They are reflex sight images from right eye view and scaled by a Nearest Neighbor interpolation method for better comparison of details at the pixel level.

We used a Simulator Sickness Questionnaire (SSQ) to measure the levels of sickness [3] and the revised Game Experience Questionnaire (GEQ) to measure the gameplay experience [2]. Both were delivered after each condition. The SSQ results were further processed to derive relative SSQ scores (Δ SSQ) with its Total Severity sub-score. While the revised GEQ had five sub-scores: flow, immersion, competence, positive affect, and negativity [2]. In addition, we also collected participants' gameplay performance in terms of the number of fired ammo and hits received in each condition.

Sixteen volunteers (6 females, 10 males) were recruited from a local university. Participants' age ranged between 18 and 22 (19.88 ± 1.11). The experiment was conducted in accordance with the ethics guidelines and regulations of the [removed for review]. A PIMAX 8K VR HMD was used in the experiment. It is a tethered VR device with a diagonal FOV of 200° , a refresh rate of 90HZ, and a resolution of 3840×2160 per eye. Although the driver for VR games (i.e., SteamVR) can set the rendering resolution above 4K, we still used 4K as the highest resolution condition, which was close to the hardware limits and represented the highest resolution available in today's commercial HMDs (see Table 1).

3 RESULTS AND DISCUSSION

In this section, we present a summary of some of the main results of our experiment. The Friedman test revealed a statistically significant difference in SSQ among the four resolution conditions ($\chi^2(2) = 17.234, p = 0.001$). Post-hoc pairwise comparisons showed significant differences when comparing the scores of 1K, 3K, and 4K (3K-1K: $Z = -2.952, p = 0.003$, 4K-1K: $Z = -2.873, p = 0.004$) in favour of 3K and 4K (lower SSQ scores). On the other hand, the results of Friedman test indicated statistically significant differences in the flow scores ($\chi^2(2) = 8.764, p = 0.033$) and positive affect scores ($\chi^2(2) = 15.205, p = 0.002$) among the different resolutions. Post-hoc analysis revealed significant differences when comparing the flow scores of 1K and 2K (2K-1K: $Z = -2.711, p = 0.007$) in favour of 2K (higher flow scores), as well as the positive affect scores of 1K, 3K, and 4K (3K-1K: $Z = -3.020, p = 0.003$, 4K-1K: $Z = -2.949, p = 0.003$) in favour of 3K and 4K (higher positive affect scores). A repeated-measure ANOVA did not yield any significant differences in the number of ammo fired and hits received among all conditions ($p = 0.450$, and 0.647 , respectively).

The results of SSQ analysis indicate that 2K, 3K, and 4K can produce significantly lower SS symptoms than 1K. The results of GEQ analysis indicate that 2K can help the participants to enter into a significantly deeper flow state than 1K. Moreover, 3K and 4K can produce a significantly stronger positive affect than 1K. However, there are no significant differences among 2K, 3K, and 4K for each

factor of both SSQ and GEQ. The analysis of participants' performance shows that the resolutions did not affect players' performance in VR FPS game significantly.

4 CONCLUSION AND FUTURE WORK

In this paper, we presented our explore of the effect of resolution on game experience and simulation sickness (reduction) in Virtual Reality (VR) environments, especially in games. We used a modified game level from *Half-Life: Alyx* as the testing environment to explore the benefits of higher resolution on SS reduction and enhanced gameplay experience. The analysis of experiment results showed that with 2K, 3K, and 4K, users had a significantly better game experience and significantly fewer SS symptoms than with 1K. These results suggest that 2K is an important threshold resolution that can provide a good gaming experience with lower SS symptoms but without any adverse effect on players' performance. Our experiment has focused on resolution and did not take into account other aspects like frame rate and field-of-view. We plan to conduct further experiments where other aspects can be relaxed and examined in a systematic way to assess their effect of simulator sickness, performance and experience. In addition, we plan to conduct a series of evaluations on other VR applications and gaming scenarios that focus on slower pace and other gameplay requirements and determine the effects of resolution and associated rendering issues.

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